25901751 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCPHY100 - MODERN PHYSICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Give the postulates of special theory of relativity. [K] / [CO1]
2. What happens to the energy of a body as it approaches the speed of light? [A] / [CO1]
3. Distinguish between inertial and non-inertial frames of reference. [U] / [CO1]
4. A red light has a wavelength of 632.8 nm. What is the frequency? [A] / [CO2]
5. How does the presence of a medium influence the propagation of electromagnetic waves? [U] / [CO2]
6. Electron microscopes have better resolution than optical microscopes. Justify the statement. [A] / [CO2]
7. In photoelectric effect blue light results in faster electrons than red light. Why? [U] / [CO2]
8. What are the features of a blackbody? [K] / [CO2]

9. Explain the origin of emission and absorption spectral lines. [K] / [CO3]

10. Explain the concept of atomic excitation in terms of electron transitions. [U] / [CO3]

11. Give the expansion of the acronym LASER. [K] / [CO4]

12. Write the expressions for energy eigen value and wave function corresponding to the lowest energy level of a particle in a one dimensional box of length L [A] / [CO5]

13. Write steady state form of Schrodinger wave equation in one dimension. [K] / [CO5]

14. What is a wave function in quantum mechanics? Explain its physical significance. [U] / [CO5]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

15. The measured life of a particle moving with a velocity 0.92 c is $3.4 \times 10^{-6} s$. Calculate its proper life. [A] / [CO1]

16. Discuss the concept of length contraction. Obtain the relation between measured length and proper length. [U] / [CO1]

17. A metal has a work function of 3 eV. What is the longest wavelength of light that can cause photoelectric emission from this metal? [A] / [CO2]

18. Find the de Broglie wavelength of a 1.0 mg grain of sand blown by the wind at a speed of 20 m/s. [A] / [CO2]

19. The position and momentum of a 1 KeV electron are simultaneously determined. If its position is located within 1 angstrom, what is the percentage of uncertainty in its momentum? [A] / [CO2]

20. Explain why the spacing between spectral lines decreases as we move towards higher quantum numbers in each hydrogen spectral series. [U] / [CO3]

21. Using a neat diagram, describe the process of stimulated absorption, spontaneous emission and stimulated emission of radiation. [U] / [CO4]

22. Using classical wave equation arrive at the time dependent Schrodinger wave equation in one dimension. [U] / [CO5]

[5x6 = 30]